CIBC HACKATHON FINALS · TEAM ADDA WALE

Collections Copilot

Every collections call in the right order, within policy, with the reason shown.

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Synthetic Maple Bank data · snapshot 28 September 2026

Before every call, an agent checks eight things

1 · Stop
Restrictions
Bankruptcy, cease request, deceased

2
Representative
Talk only to their lawyer or counsellor

3
Disputes
Pause the disputed amount

4
Vulnerability
Gentle script, certified agent

5
Hardship
Offer a programme before payment

6
Open promise
Remind; don’t ask for more

7
Legal window
Local hours, holidays, 3 tries in 7 days

8 · Only then
Payment ask
Timed to payday, within 21 days

Facts sit in five systems, rules in 19 policies. **Miss one, and it’s a breach or a harmed customer.**

What breaks today, across 96,000 open cases

760

cases contacted more often than the 7-day cap allows

1,040

contacts outside legal hours or on a holiday, last 30 days

18,758

customers in hardship with no support plan yet

Behind them: **$69.8M** on cases close to write-off. Every number is one SQL query, shown in the app.

How we decided which tool does each job

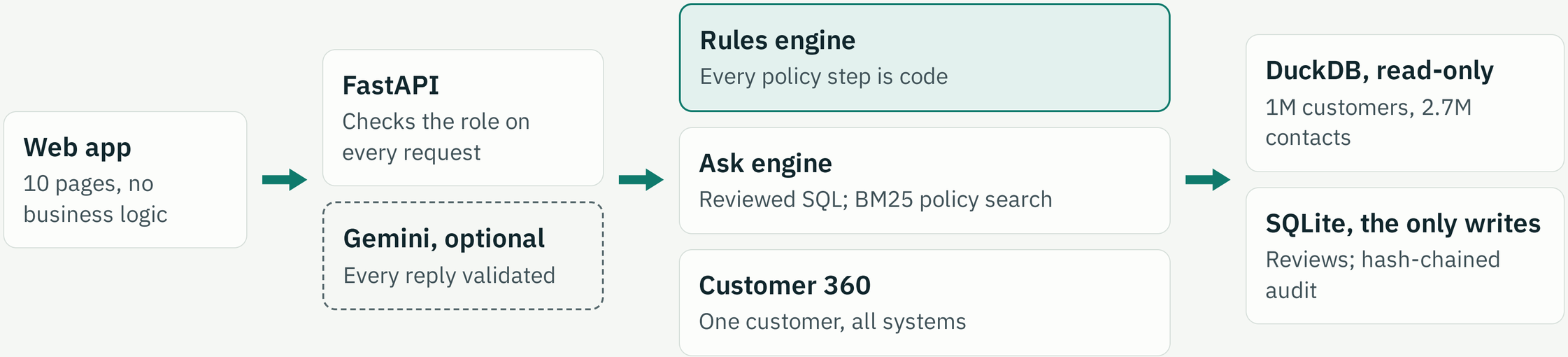
Five constraints, before any code

1. Compliance first
2. Explainable: every step cites its policy
3. Customer text stays in the bank
4. Works on the next data release
5. Every number measured, not claimed

Written in policy?	**Rules** Contact hours, the weekly cap, restrictions, eligibility
A pattern in text, with labelled examples?	**A small model** Hardship cues in agent notes
Language we can check against a source?	**LLM proposes, app verifies** Questions, evidence, call notes
None of these?	**A person decides** Eligibility, approving every brief and call

The LLM goes where language helps; it never touches numbers or compliance steps.

The backend, and why each part



ms
30-day scan of all contacts

0.17 s
Call Mode update

21 s
Plan all 93,320 customers

0
Model-written SQL ever run

Why a small model, not an LLM

Same 126 test notes	Hardship found	False alarms	Per note	Leaves the bank
Our detector	38 of 38	0	under 1 ms	No
Gemini 3.5 Flash-Lite	38 of 38	4	1.4 s	Yes

3 of its 4 false alarms

were notes tagged “cust vulnerable - financial hardship”: a vulnerability tag under policy, not hardship.

So the LLM only proposes

Gemini 3.5 Flash-Lite: fast, JSON we validate, free tier. The app verifies; a person accepts.

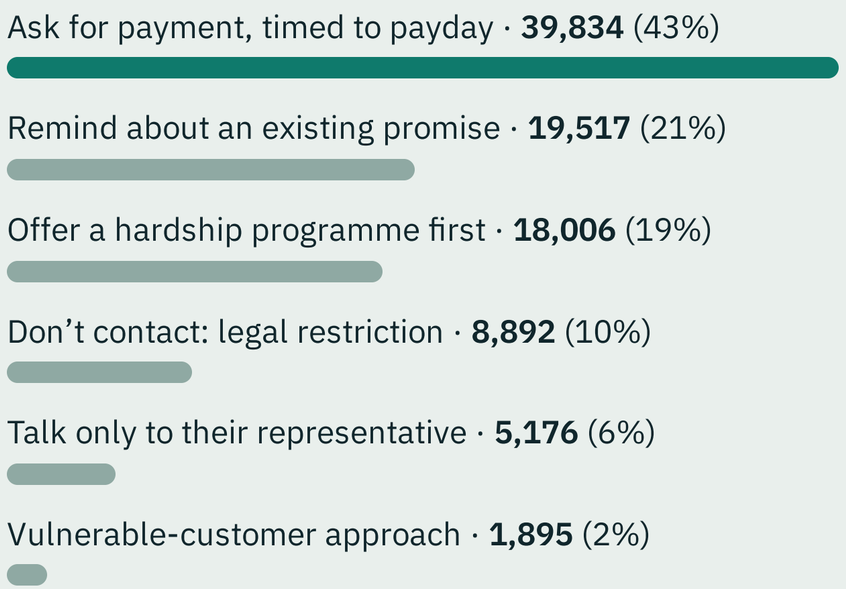
Every open case, planned every night

93,320

customers planned in 21 seconds on one laptop

0 errors · the same rules as every brief and call

Today’s first step for each customer



Fewer than half should simply be asked to pay.

Built for the next data release, not just this one

What	Read from the data	Fake March 2027 release
Today’s date	The extract’s own timestamp	24 Mar 2027 detected
Statutory holidays	Computed by rule (105 of 105)	Good Friday skipped
Weekly contact cap	Each case’s own cap	Cap of 2 blocks 2 days
Policy versions	The release’s documents	Cites the new v5.0
Programme rules	Version in effect that day	2027 terms used

No code changes.

Our hardship detector, measured honestly

Recall

68 → 95%

of hardship notes found

Precision

90 → 98%

of flags that were right

Unseen notes

36 of 38

found, with 0 false alarms

Caveat: the synthetic notes reuse phrasings, so this is coverage of this dataset, not real-world accuracy. No protected or vulnerability inputs.

Ask in plain English; every answer shows its SQL

Answer first

A one-line summary, then the table and its SQL

Clarify, don’t guess

Unsupported questions get rephrasings, never a substitute number

Refuse what’s not allowed

Personal-data exports and ranking by protected traits

12/12

organiser reference answers matched

50

benchmark answers, same engine

559

automated tests passing

TO SUM UP

Rules decide.
AI suggests.
People approve.

Limits we state up front: synthetic data · national holidays only · payday estimated for 52% of cases · the app never contacts a customer

Thank you. Questions?